Module 1: Transformer from First Principles

Unpacking a Decoder-Only LM

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Lecture map

1. Next-token prediction
2. Attention from first principles
3. The Transformer block
4. From GPT-2 to modern LLMs & Lab 1

Guiding question

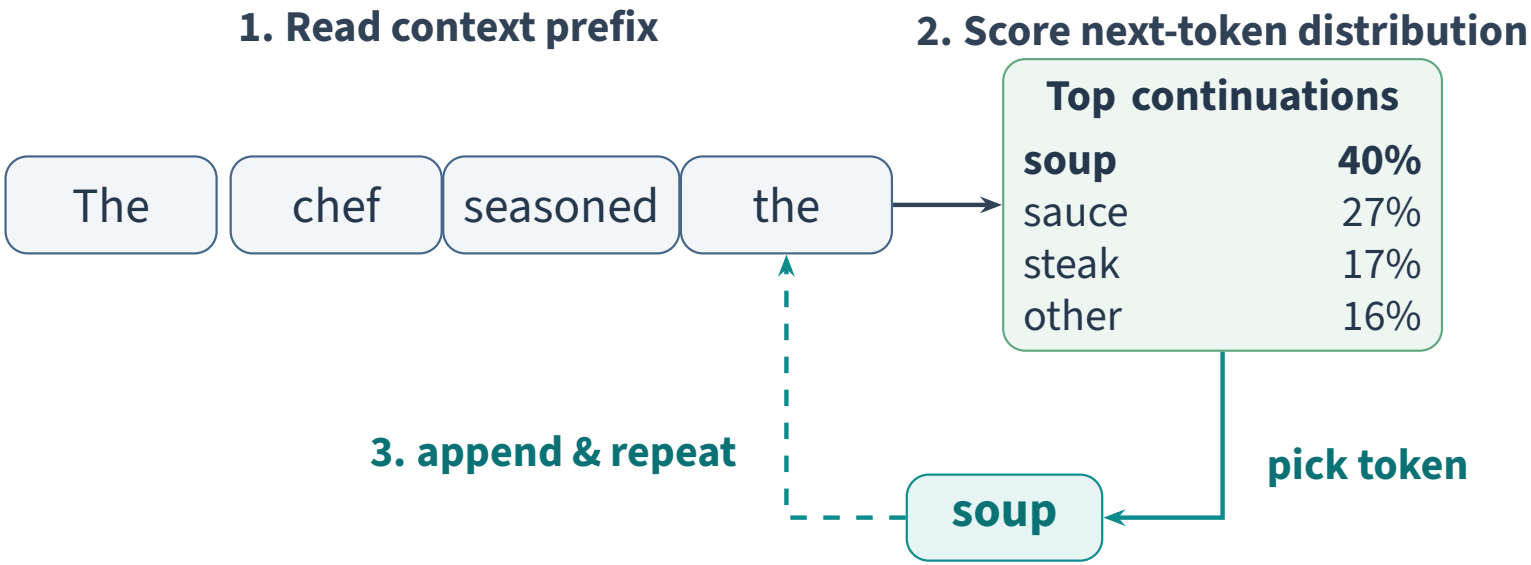
How does a decoder-only LLM generate text?

PART 1

Next-token prediction

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The task: sequence in, next-token distribution out



At every step, use the prefix to score the vocabulary $\mathcal{V}$, append one token, and repeat.

▷ **Live demo: Explore next-token prediction (next-token.html)**

Formalizing next-token prediction (NTP)

Autoregressive factorization

By the chain rule, the joint probability $P(X)$ factorizes into conditional steps:

$$P(X) = \prod_{t=1}^{T} P(x_t \mid x_{<t})$$

Modeling a text sequence is equivalent to predicting each token from prefix $x_{<t}$.

Self-supervised training loss

Natural text is self-labeled: token x_t is the target for prefix $x_{<t}$. Cross-entropy:

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_{t=1}^{T} \log P_\theta(x_t \mid x_{<t})$$

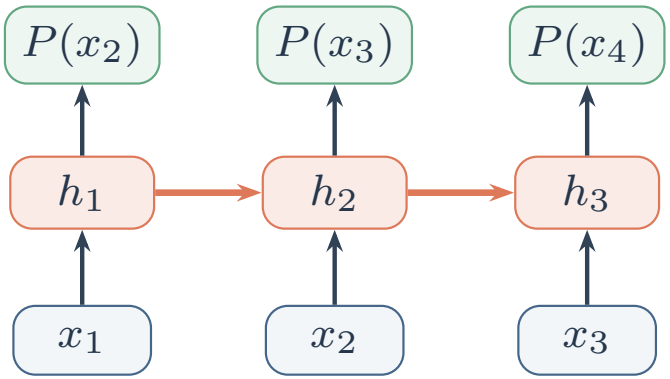
One sequence of length T provides T supervised prediction targets in parallel.

Source: Bengio et al. (2003); Radford et al. (2018, 2019)

Same task, two ways to process the sequence

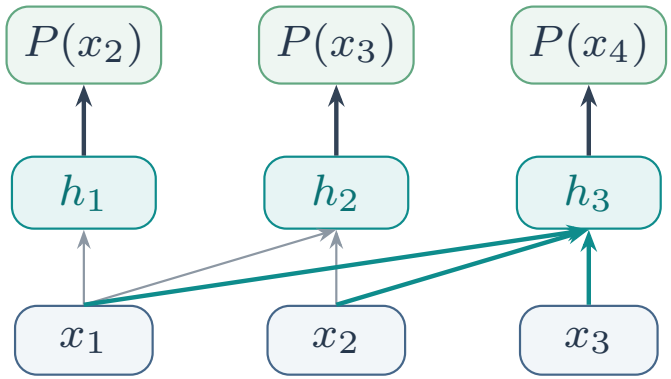
At every position t, predict $P(x_{t+1} \mid x_{\leq t})$ from the prefix so far.

Recurrent model: inherently sequential



$h_t = f(h_{t-1}, x_t)$: step t waits on step $t-1$.

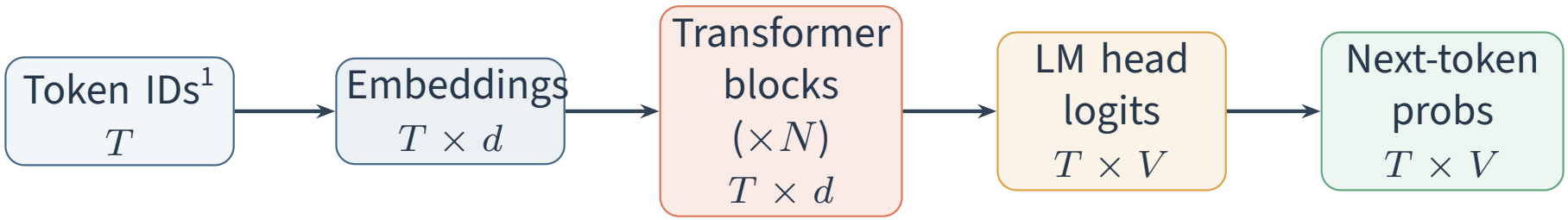
Causal Transformer: inherently parallel



No $h_{t-1} \to h_t$ wait: every position gathers $x_{\leq t}$ directly via attention in one pass.

Source: Elman (1990); Hochreiter & Schmidhuber (1997); Pascanu et al. (2013); Vaswani et al. (2017)

The whole model, at a glance



1. Embed

Token IDs index into lookup table $W_E \in \mathbb{R}^{V \times d}$ to become continuous latent vectors in $\mathbb{R}^d$.

2. Process ($\times N$)

N identical blocks alternate cross-token mixing (**attention**) and per-token reasoning (**MLP**).

3. Unembed

Linear head maps latent vectors back to vocabulary scores (logits $\in \mathbb{R}^V$); softmax yields probabilities.

Unpacking the pipeline from first principles

We now walk through each stage step by step: starting with how discrete tokens enter continuous latent space (**Step 1: Embed**), then how **causal attention** allows them to communicate (**Steps 2–5**).

[1] For now, 1 token = 1 word.

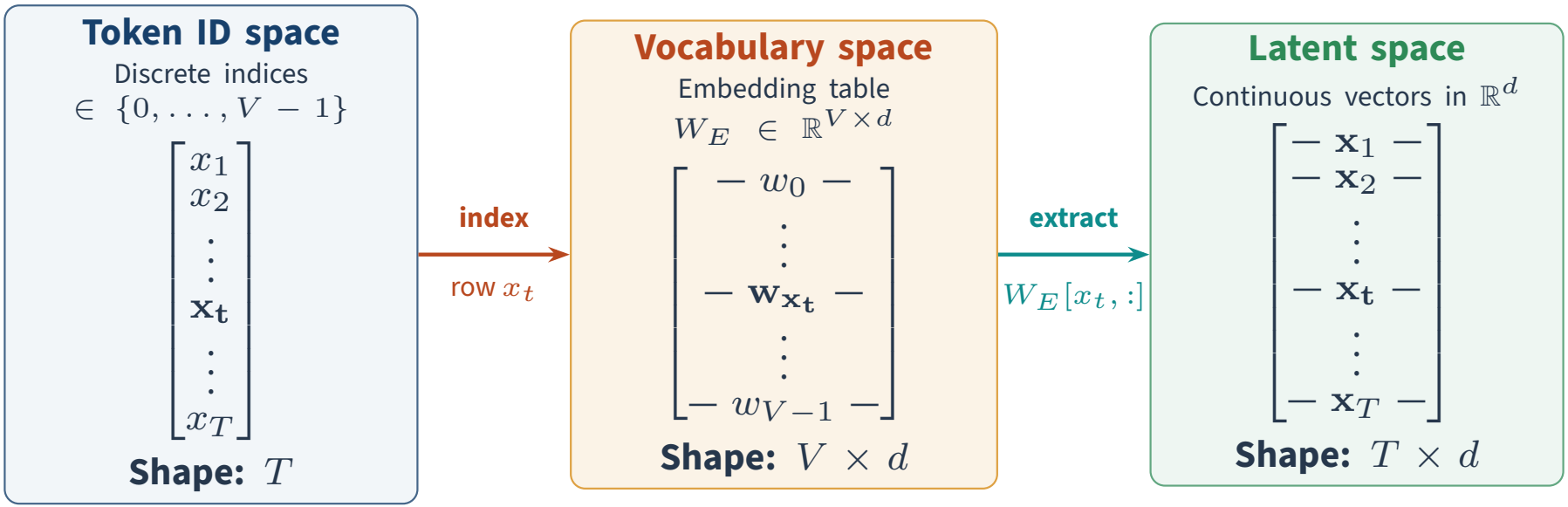
PART 2

Attention from first principles

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Step 1: Tokens become vectors in a shared space

Stage 1 (Embed): Mapping discrete tokens into continuous latent vectors before Attention:



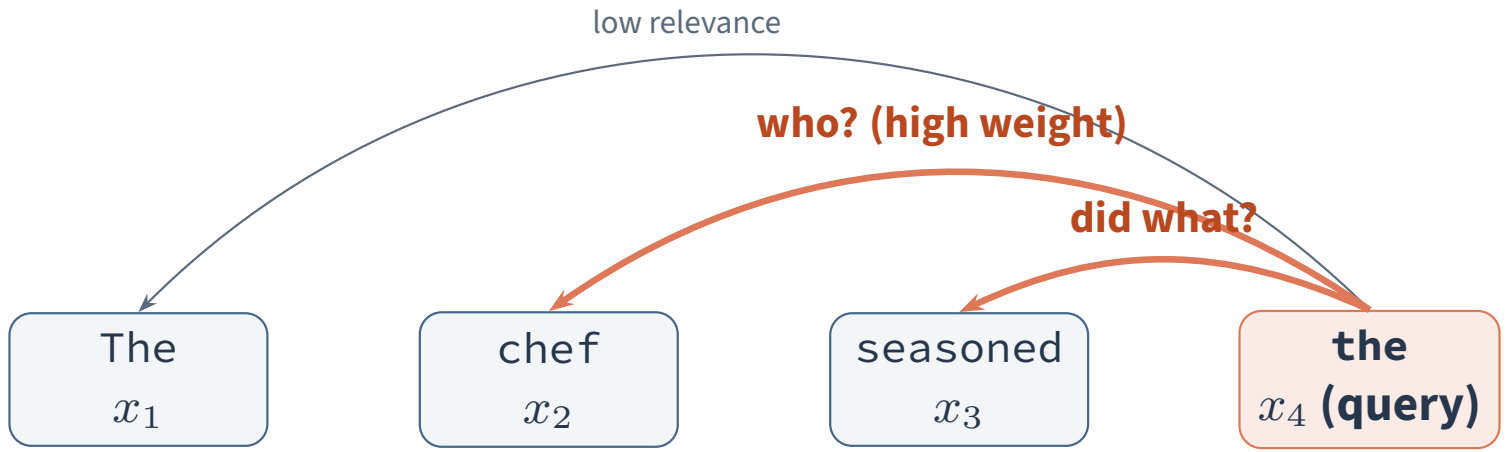
Why initial embeddings fail: context-blind vectors

Row $\mathbf{x}_t = W_E[x_t]$ is a static dictionary lookup. Positions 1 and 4 ("the") get identical vectors with zero context.

$\Longrightarrow$ **Tokens must communicate across the sequence: this is Attention (Step 2).**

Step 2: Relating tokens across the sequence

To predict what follows **“the”**, its representation must pull context from related earlier tokens:



The goal of self-attention

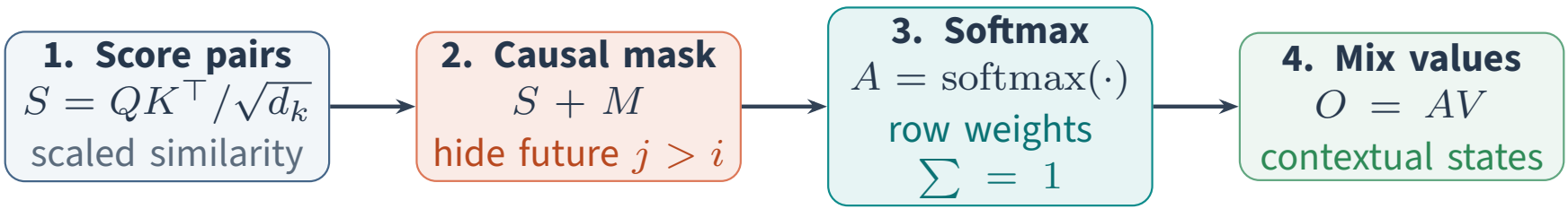
For every token i, compute a relevance weight $\alpha_{i,j} \geq 0$ ($\sum_{j \leq i} \alpha_{i,j} = 1$) for each allowed token $j \leq i$, and form a weighted mixture of their information.

Source: Bahdanau et al. (2014); Vaswani et al. (2017)

Step 3: Scaled dot-product attention

The core engine of the Transformer takes Query, Key, and Value matrices $Q, K, V \in \mathbb{R}^{T \times d_k}$:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V$$



Why divide by $\sqrt{d_k}$? Preserving gradient flow

For independent zero-mean unit-variance components, $\text{Var}(q^\top k) = d_k$. As dimension d_k grows, large dot products push softmax into saturated flat regions with vanishing gradients ($\nabla \approx 0$). Dividing by $\sqrt{d_k}$ keeps $\text{Var}(S) \approx 1$, maintaining healthy gradient flow.

Source: Vaswani et al. (2017, *Attention Is All You Need*)

Step 4: The intuition of Queries, Keys, and Values

Why project X into learned Q, K, V spaces instead of computing $\text{softmax}(XX^\top/\sqrt{d})X$?

Query ($q_t = x_t W_Q$)
What am I looking for?
Search criteria / prompt

Key ($k_t = x_t W_K$)
What features do I have?
Identity / catalog index

Value ($v_t = x_t W_V$)
What content do I transmit?
Content payload

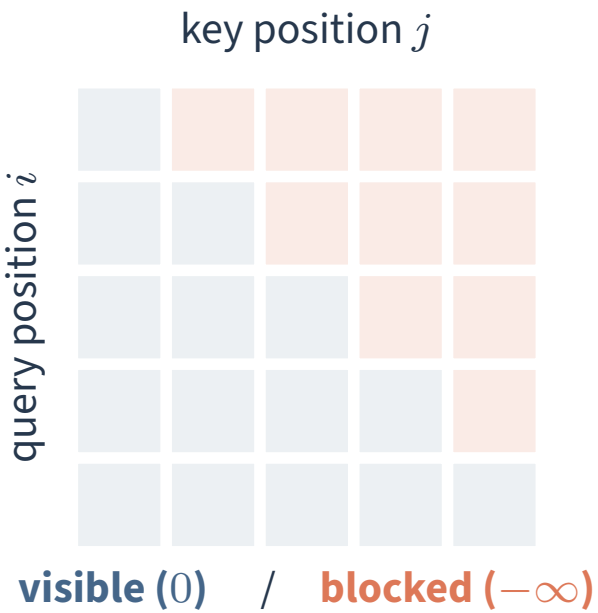
Search Engine Analogy

- **Query**: search prompt entered by user.
- **Keys**: webpage titles and index tags.
- **Values**: actual page text returned.

Why separate projections?

- **Asymmetry**: $x_i^\top x_j$ is symmetric, but language relations are directed (verb $\to$ noun $\neq$ noun $\to$ verb).
- **Decoupling**: what matches a token differs from the information it delivers.

The causal mask is a constraint on logits



- Position i may attend only to keys $j \leq i$.
- Future scores $(j > i)$ receive $-\infty$ **before** softmax so $\exp(-\infty) = 0$.
- Zeroing probabilities after softmax breaks row normalization.

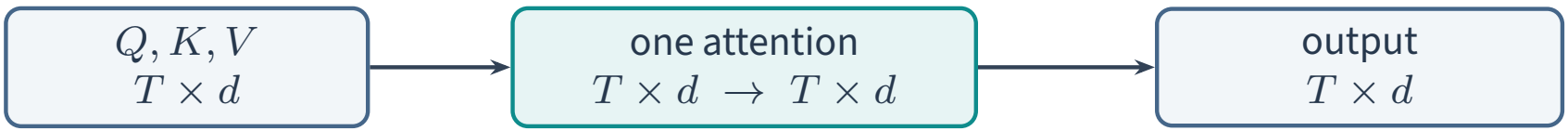
Checkpoint: one query row

Think, then compare

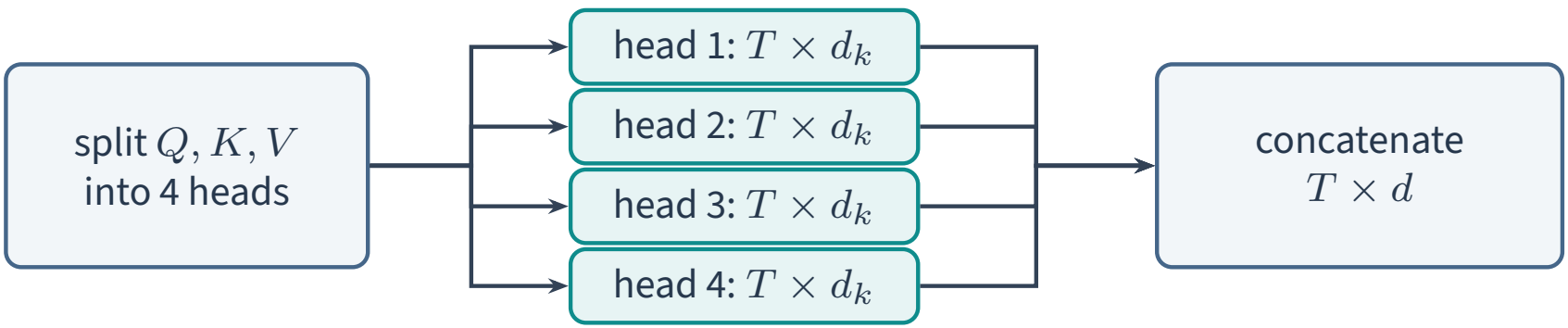
For sequence length four, write the four logits seen by the query at position two (0-indexed). Which entries survive the mask, and across which dimension must softmax normalize?

One wide head or several smaller heads?

One head: width d



Multi-head example: $h = 4$ heads, each width $d_k = d/4$



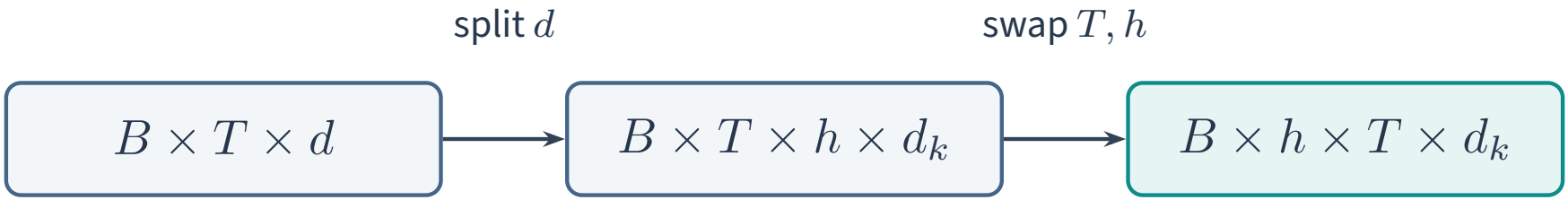
Same leading attention compute, different mixing

Per batch: one head $O(BT^2d)$; h heads $O(BhT^2d_k) = O(BT^2d)$. Heads have learned projections and can compute different attention patterns in parallel.

Each head receives Q, K and V from learned projections; W_O mixes the concatenated heads.

Split and swap: give each head a whole sequence

For each projected tensor Q, K, V, make the head dimension explicit:



Each head now sees a $T \times d_k$ sequence, so batched $QK^\top$ gives $B \times h \times T \times T$ scores. Heads can run concurrently.

Concrete example

$d = 768, h = 12$: split 768 into 12×64, then swap the token and head axes. Reverse the swap and concatenate the heads after attention.

In PyTorch: `reshape(B, T, h, d_k).transpose(1, 2)`.

What does multi-head attention cost?

For B sequences of T tokens, width d, and $d_k = d/h$ (with h dividing d):

Forward operation	One width-d head	h width-d_k heads
Score and value products	$4BT^2d$	$4BhT^2d_k = 4BT^2d$
Q, K, V and output projections	$8BTd^2$	$8BTd^2$
Score elements, if materialized	BT^2	BhT^2

Interpretation

Splitting the fixed width does not multiply the leading matrix-product FLOPs by h. Multiple heads learn different attention patterns; parallel execution is possible, but latency depends on kernels and hardware.

Dense products; two FLOPs per multiply-add. Excludes masking, softmax and biases. Causal kernels may skip work and avoid materializing scores.

What does the causal mask reveal about order?

Without a mask or positions

Self-attention is **permutation equivariant**: reordering the input reorders its outputs in the same way.

With a causal mask

Each token sees a different prefix. Reordering tokens changes who can see whom, even without explicit positional embeddings.

Sequence	History visible to A	History visible to B
$A\,B\,C$	A	A, B
$B\,A\,C$	B, A	B

Earlier token representations can therefore differ; a later layer can use them to distinguish the two orders.

Source: Haviv et al. (2022); Irie (2025)

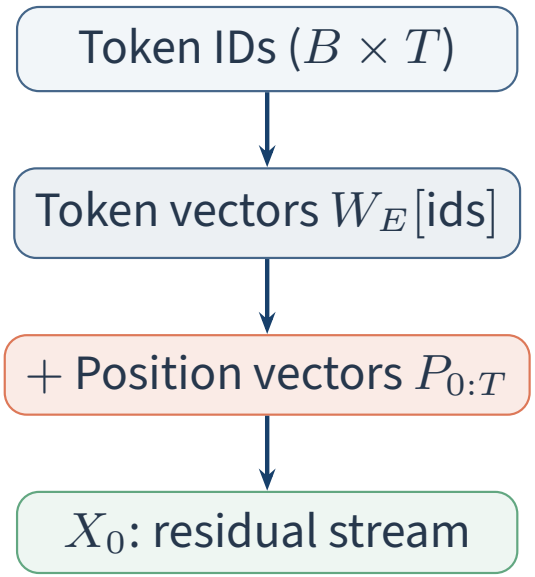
Explicit position information in GPT-2 and beyond

Add an explicit positional signal

GPT-2 adds learned position vectors to token embeddings:

$$X_0 = W_E[\text{ids}] + P_{0:T}.$$

Many modern LLMs apply RoPE to Q and K instead.



The causal mask itself also provides order information.

Original Transformer: fixed sinusoidal positions

For position p and feature pair $2i, 2i+1$ ($d = d_{\text{model}}$):

$$\text{PE}(p, 2i) = \sin\left(\frac{p}{10000^{2i/d}}\right), \qquad \text{PE}(p, 2i+1) = \cos\left(\frac{p}{10000^{2i/d}}\right).$$

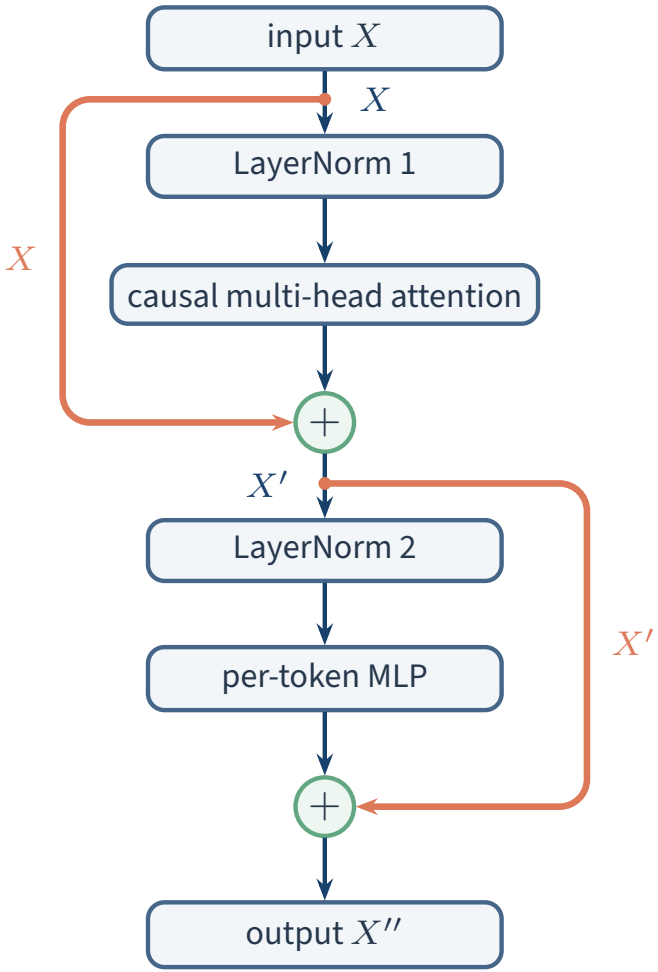
Source: Vaswani et al. (2017); Radford et al. (2019); Su et al. (2021)

PART 3

The Transformer block

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A pre-norm Transformer block has two residual updates



First residual update

$$X' = X + \mathrm{MHA}(\mathrm{LN}_1(X))$$

Attention mixes information across token positions.

Second residual update

$$X'' = X' + \mathrm{MLP}(\mathrm{LN}_2(X'))$$

The MLP transforms each token independently.

The orange paths carry the unchanged stream around each sublayer. Every addition preserves $B \times T \times d$.

Source: He et al. (2016); Radford et al. (2019); Xiong et al. (2020)

Inside the MLP and the $1/3$ vs. $2/3$ parameter rule

The MLP applies a two-layer feed-forward network to each token vector $x_t \in \mathbb{R}^d$ independently:

$$\mathrm{MLP}(x_t) = \mathrm{GELU}(x_t W_1 + b_1) W_2 + b_2$$

- **Expand**: $W_1 \in \mathbb{R}^{d \times 4d}$ projects $d \to 4d$.
- **Activate**: elementwise GELU non-linearity.
- **Contract**: $W_2 \in \mathbb{R}^{4d \times d}$ projects $4d \to d$.

Parameter budget per block

- **MHA** (W_Q, W_K, W_V, W_O): $\mathbf{4d^2}$ **(1/3)**
- **MLP** (W_1, W_2): $\mathbf{8d^2}$ **(2/3)**
- **Total per block**: $\mathbf{12d^2}$ parameters

Takeaway for scaling

Two-thirds of block parameters and matrix-multiply FLOPs live in the MLP!

Source: Vaswani et al. (2017); Hendrycks & Gimpel (2016, GELU)

Normalization: stabilizing activations and gradients

LayerNorm (per token across d)

For each token vector $x \in \mathbb{R}^d$:

$$\mathrm{LN}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \odot \gamma + \beta$$

Centers to zero mean (μ), scales to unit variance (σ^2), then applies learned $\gamma, \beta \in \mathbb{R}^d$.

Pre-Norm vs. Post-Norm

- **Pre-Norm (GPT-2 / modern LLMs)**: $X + F(\mathrm{Norm}(X))$. Identity highway bypasses norm; stable gradients at initialization.
- **Post-Norm (2017 Transformer)**: $\mathrm{Norm}(X + F(X))$. Requires careful warmup; different model!

Modern speedup: RMSNorm drops mean-centering (μ) and β, normalizing only by $\sqrt{\frac{1}{d}\sum_i x_i^2 + \epsilon}$.

Source: Ba et al. (2016, LayerNorm); Zhang & Sennrich (2019, RMSNorm); Xiong et al. (2020)

Closing the loop: final norm, LM head, and target shift

After N Transformer blocks, normalize X_N and project to vocabulary logits:

$$Z = \mathrm{Norm}_f(X_N) W_{\text{vocab}}^\top \qquad \in \mathbb{R}^{B \times T \times V}$$

$$\mathcal{L} = -\frac{1}{BT} \sum_{b,t} \log \mathrm{softmax}(Z_{b,t})_{y_{b,t}}$$

- **Weight tying (GPT-2)**: shares $W_{\text{vocab}} = W_E \in \mathbb{R}^{V \times d}$, saving $V \times d$ parameters (≈ 38.6M in GPT-2 Small).

One stream, two views

```
x = tokens[:, :-1]
y = tokens[:, 1:]
```

Common bug

Forgetting the $+1$ shift trains the model to copy the current token ($x_t \to x_t$): loss drops to zero, generation fails completely.

Source: Press & Wolf (2017, Weight Tying); Radford et al. (2019)

Interactive walkthrough: visual explorer of the Transformer

Session cue — trace a live forward pass:

1. Follow **token + position embeddings** into the residual stream.
2. Zoom into the **Transformer block**: inspect Q, K, V projections, causal attention weights, and head concatenation.
3. Trace the **MLP expansion/contraction** and final **LM head** to next-token logits.

Visual Explorer Demo

Open the interactive canvas and hover over each component to inspect tensor shapes and data flow:

▷ **Open decoder-transformer.html**

See also: Transformer Explainer (GPT-2 live)

PART 4

From GPT-2 to modern LLMs & Lab 1

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Three Transformer families—and why Decoder-Only won

Encoder-only

BERT (Devlin et al., 2018)

Bidirectional attention for classification and search. No autoregressive generation.

Encoder–decoder

Original / T5 (Raffel, 2020)

Bidirectional source encoder + causal target decoder via cross-attention.

Decoder-only

GPT / LLaMA (Radford, 2019)

Single causal stream unifies all tasks as prompt completion + single KV cache.

Why decoder-only dominates modern foundation models

Any task can be framed as prompt continuation, concentrating all parameters and pretraining compute into one unified causal stack with a single KV cache.

Source: Vaswani et al. (2017); Devlin et al. (2018); Radford et al. (2019); Brown et al. (2020)

What changed since GPT-2? The modern LLM stack

You will implement **GPT-2** in Lab 1. Modern LLMs keep the same macro-architecture while upgrading five components:

Component	GPT-2 (2019)	Modern LLM (2023+)	Primary rationale
Positions	Learned W_{pe}	**RoPE** (Su et al., 2021)	Relative distance, length extrapolation
Normalization	Pre-**LayerNorm**	Pre-**RMSNorm** (Zhang, 2019)	Drops mean-centering; 10–50% faster
Attention	**MHA** ($H_Q = H_{KV}$)	**GQA** (Ainslie et al., 2023)	$4\times$–$8\times$ smaller KV cache in serving
MLP activation	**GELU** ($d \to 4d \to d$)	**SwiGLU** (Shazeer, 2020)	Gated linear unit improves capacity
Biases & tying	Biases; tied W_E	**Bias-free**; untied W_{vocab}	Faster GEMMs; better stability at scale

Source: Touvron et al. (2023, LLaMA); Jiang et al. (2023, Mistral); Raschka (LLM Architecture Gallery)

A shape ledger catches semantic bugs

Tensor	Shape	Role in `nanolm/model.py`
`idx`	$B \times T$	input token IDs ($T \leq$ block_size)
`x`	$B \times T \times d$	residual stream (token + position)
`q, k, v`	$B \times H \times T \times d_h$	per-head projections ($d_h = d/H$)
`att`	$B \times H \times T \times T$	masked attention weights (rows sum to 1)
`mlp_hidden`	$B \times T \times 4d$	expanded per-token MLP features
`logits`	$B \times T \times V$	next-token vocabulary scores

Review question

Over which axis of `att` ($B \times H \times T \times T$) does `softmax` normalize, and why must masking happen before it?

Lab 1: what you have the tools to build and verify now

Implement in `nanolm/model.py`

1. **Session 2:** embeddings, MLP and residual-block sketch.
2. **Session 3:** Q/K/V projections, head split, causal attention and the full block.
3. **Session 4:** assemble `MiniGPT`, test causality and compare GPT-2 logits.

Three verification gates

1. **Shape tests**: all intermediates match the ledger.
2. **Causality tests**: mutating a future token leaves earlier logits identical.
3. **GPT-2 parity**: compare reference logits. Cache the 124M checkpoint beforehand; if unavailable, report parity as pending.

`uv run pytest tests/test_module_01_model.py -q` → `uv run --extra gpt2 python scripts/01_gpt2_parity.py`

Checkpoint: debugging order

Think, then compare

Your model passes all shape tests, but `01_gpt2_parity.py` produces wrong logits and the causality test fails. Which lines of `CausalSelfAttention.forward` do you inspect first?

Key takeaways

1. Next-token prediction factorizes sequence probability; removing the $h_{t-1} \to h_t$ dependency makes training inherently parallel across positions on GPUs.
2. Causal multi-head attention routes information across tokens ($1/3$ of params), while the causal mask and position representations supply order information.
3. A pre-norm Transformer block pairs cross-token attention with a per-token $4d$ MLP ($2/3$ of params) along a clean residual highway.
4. Shape invariants, causality perturbation tests, and exact GPT-2 checkpoint parity verify the model before we train it in Module 2.